

Chapter 5

MATRICES

Real-life Motivation

Matrices play an important role in modern mathematics and its applications. They are used to organize data, solve systems of linear equations, represent transformations in geometry, and model real-life situations in physics, engineering, economics, statistics, and computer science. Mastering matrices is essential for higher studies in mathematics and applied sciences.

5.1 Introduction to Matrices

Matrices are mathematical tools used to arrange numbers, symbols, or expressions in rows and columns. They provide a compact and systematic way to represent and manipulate data and equations.

Definition of a Matrix (General Form)

A **matrix** is a rectangular array of elements arranged in rows and columns and enclosed within square brackets $[\]$.

If a matrix has m rows and n columns, then it is said to be of **order** $m \times n$.

The general form of a matrix A of order $m \times n$ is:

$$A = [a_{ij}]_{m \times n} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}$$

Here, a_{ij} denotes the element in the i^{th} row and j^{th} column.

Order of a Matrix

The **order of a matrix** is defined as the **number of rows and number of columns** present in the matrix.

If a matrix has m rows and n columns, then its order is written as

$$m \times n$$

Example:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Matrix A has 2 rows and 3 columns. Hence, the order of A is 2×3 .

5.2 Types of Matrices

Row Matrix

A **row matrix** is a matrix that has **only one row**.

Example:

$$A = \begin{bmatrix} 2 & 5 & 7 \end{bmatrix}$$

This is a row matrix of order 1×3 .

Column Matrix

A **column matrix** is a matrix that has **only one column**.

Example:

$$B = \begin{bmatrix} 4 \\ 6 \\ 9 \end{bmatrix}$$

This is a column matrix of order 3×1 .

Rectangular Matrix

A **rectangular matrix** is a matrix in which the **number of rows is not equal to the number of columns**.

Example:

$$C = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

This is a rectangular matrix of order 2×3 .

Square Matrix

A **square matrix** is a matrix in which the **number of rows equals the number of columns**.

Example:

$$D = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

This is a square matrix of order 2×2 .

Zero Matrix

A **zero matrix** is a matrix in which **all elements are zero**.

Example:

$$O = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Diagonal Matrix

A **diagonal matrix** is a square matrix in which **all non-diagonal elements are zero**.

Example:

$$E = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 7 \end{bmatrix}$$

Scalar Matrix

A **scalar matrix** is a diagonal matrix in which **all diagonal elements are equal**.

Example:

$$F = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

Identity Matrix

An **identity matrix** is a scalar matrix in which **each diagonal element is 1**.

Example:

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Symmetric Matrix

A square matrix A is called **symmetric** if

$$A^T = A$$

Example:

$$A = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$$

Skew-Symmetric Matrix

A square matrix A is called **skew-symmetric** if

$$A^T = -A$$

Example:

$$A = \begin{bmatrix} 0 & 4 \\ -4 & 0 \end{bmatrix}$$

Solved Example 1

Question: If a matrix has 18 elements, what are the possible orders it can have? What if it has 5 elements?

Solution:

Number of elements in a matrix of order $m \times n$ is $m \times n$.

For 18 elements:

$$18 = 1 \times 18, 2 \times 9, 3 \times 6, 6 \times 3, 9 \times 2, 18 \times 1$$

Hence, possible orders are:

$$1 \times 18, 2 \times 9, 3 \times 6, 6 \times 3, 9 \times 2, 18 \times 1$$

For 5 elements:

$$5 = 1 \times 5, 5 \times 1$$

Hence, possible orders are:

$$1 \times 5, 5 \times 1$$

Solved Example 2

Question: Construct a 2×2 matrix $A = [a_{ij}]$ where

$$a_{ij} = \frac{(i+j)^2}{2}$$

Solution:

For a 2×2 matrix, $i, j = 1, 2$.

$$a_{11} = \frac{(1+1)^2}{2} = 2,$$

$$a_{12} = \frac{(1+2)^2}{2} = \frac{9}{2},$$

$$a_{21} = \frac{(2+1)^2}{2} = \frac{9}{2},$$

$$a_{22} = \frac{(2+2)^2}{2} = 8.$$

$$A = \begin{bmatrix} 2 & \frac{9}{2} \\ \frac{9}{2} & 8 \end{bmatrix}$$

Solved Example 3

Question: Construct a 3×4 matrix $A = [a_{ij}]$ where

$$a_{ij} = 2i - j$$

Solution:

Here $i = 1, 2, 3$ and $j = 1, 2, 3, 4$.

$$A = \begin{bmatrix} 1 & 0 & -1 & -2 \\ 3 & 2 & 1 & 0 \\ 5 & 4 & 3 & 2 \end{bmatrix}$$

Solved Example 4

Question: Find the values of x , y and z from:

$$\begin{bmatrix} x + y + z \\ x + z \\ y + z \end{bmatrix} = \begin{bmatrix} 9 \\ 5 \\ 7 \end{bmatrix}$$

Solution:

Equating corresponding elements:

$$x + y + z = 9 \quad (1)$$

$$x + z = 5 \quad (2)$$

$$y + z = 7 \quad (3)$$

From (1) – (2):

$$y = 4$$

Substitute $y = 4$ in (3):

$$4 + z = 7 \Rightarrow z = 3$$

Substitute $z = 3$ in (2):

$$x + 3 = 5 \Rightarrow x = 2$$

$$\boxed{x = 2, y = 4, z = 3}$$

3.4 Operations on Matrices

Definition

In this section, we introduce the basic operations on matrices, namely:

- Addition of matrices
- Multiplication of a matrix by a scalar
- Difference of matrices

3.4.1 Addition of Matrices

Definition

If $A = [a_{ij}]$ and $B = [b_{ij}]$ are two matrices of the same order $m \times n$, then their sum is defined as

$$A + B = [a_{ij} + b_{ij}]$$

Addition is defined only when matrices are of the same order.

Example

Production of Shoes

Two factories A and B produce shoes for boys and girls in three categories:

$$A = \begin{bmatrix} 80 & 60 \\ 75 & 65 \\ 90 & 85 \end{bmatrix}, \quad B = \begin{bmatrix} 90 & 50 \\ 70 & 55 \\ 75 & 75 \end{bmatrix}$$

Find the total production.

Solution

$$A + B = \begin{bmatrix} 80 + 90 & 60 + 50 \\ 75 + 70 & 65 + 55 \\ 90 + 75 & 85 + 75 \end{bmatrix} = \begin{bmatrix} 170 & 110 \\ 145 & 120 \\ 165 & 160 \end{bmatrix}$$

Example

If

$$A = \begin{bmatrix} \sqrt{3} & 1 & -1 \\ 2 & 3 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & \sqrt{5} & 1 \\ -2 & 3 & \frac{1}{2} \end{bmatrix}$$

find $A + B$.

Solution

$$A + B = \begin{bmatrix} 2 + \sqrt{3} & 1 + \sqrt{5} & 0 \\ 0 & 6 & \frac{1}{2} \end{bmatrix}$$

Note

If two matrices are not of the same order, their sum is not defined.

3.4.2 Multiplication of a Matrix by a Scalar

Definition

If $A = [a_{ij}]$ is a matrix and k is a scalar, then

$$kA = [ka_{ij}]$$

Example

If

$$A = \begin{bmatrix} 8 & 0 \\ 4 & -2 \\ 3 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & -2 \\ 4 & 2 \\ -5 & 1 \end{bmatrix}$$

find matrix X such that $2A + 3X = 5B$.

Solution

$$2A + 3X = 5B \Rightarrow 3X = 5B - 2A$$

$$X = \frac{1}{3}(5B - 2A)$$

3.4.3 Difference of Matrices

Definition

If $A = [a_{ij}]$ and $B = [b_{ij}]$ are matrices of the same order, then

$$A - B = [a_{ij} - b_{ij}]$$

Example

If

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & -1 & 3 \\ -1 & 0 & 2 \end{bmatrix}$$

find $2A - B$.

Solution

$$2A - B = \begin{bmatrix} 2 & 4 & 6 \\ 4 & 6 & 2 \end{bmatrix} - \begin{bmatrix} 3 & -1 & 3 \\ -1 & 0 & 2 \end{bmatrix} = \begin{bmatrix} -1 & 5 & 3 \\ 5 & 6 & 0 \end{bmatrix}$$

3.4.4 Properties of Matrix Addition

Definition

For matrices A, B, C of the same order:

1. **Commutative:** $A + B = B + A$
2. **Associative:** $(A + B) + C = A + (B + C)$
3. **Additive Identity:** $A + O = A$
4. **Additive Inverse:** $A + (-A) = O$

3.4.5 Properties of Scalar Multiplication

Definition

If k, l are scalars and A, B are matrices of the same order:

1. $k(A + B) = kA + kB$
2. $(k + l)A = kA + lA$
3. $k(lA) = (kl)A$

Solved Equation Example

Example

Find the values of x and y from:

$$2 \begin{bmatrix} x & 5 \\ 7 & y - 3 \end{bmatrix} + \begin{bmatrix} 3 & -4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$$

Solution

$$\begin{bmatrix} 2x + 3 & 10 - 4 \\ 14 + 1 & 2y - 6 + 2 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$$

Comparing,

$$2x + 3 = 7 \Rightarrow x = 2, \quad 2y - 4 = 14 \Rightarrow y = 9$$

3.5 Multiplication of Matrices

Definition

Let $A = [a_{ij}]$ be a matrix of order $m \times n$ and $B = [b_{jk}]$ be a matrix of order $n \times p$. Then the product AB is defined and is a matrix of order $m \times p$, where

$$(AB)_{ik} = \sum_{j=1}^n a_{ij}b_{jk}$$

Note

Matrix multiplication is defined only when the number of columns of the first matrix equals the number of rows of the second matrix.

Method of Multiplication

Definition

To find an element in the i th row and k th column of AB :

- Take the i th row of A
- Take the k th column of B
- Multiply corresponding elements and add them

Solved Examples

Example

If

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

find AB .

Solution

$$AB = \begin{bmatrix} 1(5) + 2(7) & 1(6) + 2(8) \\ 3(5) + 4(7) & 3(6) + 4(8) \end{bmatrix} = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

Example

If

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 0 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 4 \\ 2 & -1 \\ 3 & 0 \end{bmatrix}$$

find AB .

Solution

$$AB = \begin{bmatrix} 2(1) + (-1)2 + 3(3) & 2(4) + (-1)(-1) + 3(0) \\ 1(1) + 0(2) + 2(3) & 1(4) + 0(-1) + 2(0) \end{bmatrix}$$

$$AB = \begin{bmatrix} 9 & 9 \\ 7 & 4 \end{bmatrix}$$

Non-Commutativity of Matrix Multiplication

Definition

In general,

$$AB \neq BA$$

even when both products are defined.

Example

For

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$$

find AB and BA .

Solution

$$AB = \begin{bmatrix} 4 & 6 \\ 1 & 3 \end{bmatrix}, \quad BA = \begin{bmatrix} 2 & 4 \\ 1 & 5 \end{bmatrix}$$

Hence, $AB \neq BA$.

Properties of Matrix Multiplication

Definition

For matrices A, B, C of compatible orders:

1. **Associative:** $(AB)C = A(BC)$
2. **Distributive:** $A(B + C) = AB + AC$
3. **Distributive:** $(A + B)C = AC + BC$
4. **Not Commutative:** $AB \neq BA$

Multiplication with Identity Matrix

Definition

If I is an identity matrix of suitable order, then

$$AI = IA = A$$

Zero Matrix Property

Definition

If O is a zero matrix of suitable order, then

$$AO = OA = O$$

Question

If

$$A = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 2 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 2 \\ 3 & -1 & 1 \end{bmatrix}$$

find $(AB)C$ and $A(BC)$ and hence verify that

$$(AB)C = A(BC).$$

Solution

Step 1: Find AB

$$AB = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 1 & 3 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 2 \end{bmatrix}$$
$$AB = \begin{bmatrix} 2 & -1 \\ 1 & 6 \end{bmatrix}$$

Step 2: Find $(AB)C$

$$(AB)C = \begin{bmatrix} 2 & -1 \\ 1 & 6 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 3 & -1 & 1 \end{bmatrix}$$
$$(AB)C = \begin{bmatrix} -1 & 1 & 3 \\ 19 & -6 & 8 \end{bmatrix}$$

Step 3: Find BC

$$BC = \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 3 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 5 & -1 & 5 \\ -3 & 1 & -1 \\ 7 & -2 & 4 \end{bmatrix}$$

Step 4: Find $A(BC)$

$$A(BC) = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 1 & 3 \end{bmatrix} \begin{bmatrix} 5 & -1 & 5 \\ -3 & 1 & -1 \\ 7 & -2 & 4 \end{bmatrix}$$
$$A(BC) = \begin{bmatrix} -1 & 1 & 3 \\ 19 & -6 & 8 \end{bmatrix}$$

Result

Since

$$(AB)C = A(BC) = \begin{bmatrix} -1 & 1 & 3 \\ 19 & -6 & 8 \end{bmatrix},$$

the associative law of matrix multiplication is verified.

Solved Questions

Question

Q6. Simplify:

$$\cos \theta \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} + \sin \theta \begin{bmatrix} \sin \theta & \cos \theta \\ -\cos \theta & \sin \theta \end{bmatrix}$$

Solution

$$= \begin{bmatrix} \cos^2 \theta + \sin^2 \theta & \cos \theta \sin \theta + \sin \theta \cos \theta \\ -\cos \theta \sin \theta - \sin \theta \cos \theta & \cos^2 \theta + \sin^2 \theta \end{bmatrix}$$

Using $\cos^2 \theta + \sin^2 \theta = 1$,

$$= \begin{bmatrix} 1 & 2 \sin \theta \cos \theta \\ -2 \sin \theta \cos \theta & 1 \end{bmatrix}$$

Question

Q7(i). Find matrices X and Y , if

$$X + Y = \begin{bmatrix} 6 & 2 \\ 4 & 8 \end{bmatrix}, \quad X - Y = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}$$

Solution

Adding the equations,

$$2X = \begin{bmatrix} 8 & 2 \\ 4 & 12 \end{bmatrix} \Rightarrow X = \begin{bmatrix} 4 & 1 \\ 2 & 6 \end{bmatrix}$$

Subtracting,

$$2Y = \begin{bmatrix} 4 & 2 \\ 4 & 4 \end{bmatrix} \Rightarrow Y = \begin{bmatrix} 2 & 1 \\ 2 & 2 \end{bmatrix}$$

Question

Q7(ii). Find X and Y , if

$$2X + 3Y = \begin{bmatrix} 5 & 7 \\ 1 & 9 \end{bmatrix}, \quad 3X + 2Y = \begin{bmatrix} 7 & 5 \\ 3 & 11 \end{bmatrix}$$

Solution

Subtracting the equations,

$$X - Y = \begin{bmatrix} 2 & -2 \\ 2 & 2 \end{bmatrix}$$

Solving simultaneously,

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}, \quad Y = \begin{bmatrix} -1 & 3 \\ -1 & 1 \end{bmatrix}$$

Question

Q8. Find matrix X , if

$$Y = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}, \quad 3X + Y = \begin{bmatrix} 10 & 7 \\ 8 & 9 \end{bmatrix}$$

Solution

$$3X = \begin{bmatrix} 6 & 6 \\ 6 & 6 \end{bmatrix} \Rightarrow X = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

Question

Q9. Find x and y , if

$$2 \begin{bmatrix} 1 & x \\ 0 & 2 \end{bmatrix} + \begin{bmatrix} y & 1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 6 & 5 \\ 2 & 7 \end{bmatrix}$$

Solution

Comparing corresponding elements:

$$2 + y = 6 \Rightarrow y = 4$$

$$2x + 1 = 5 \Rightarrow x = 2$$

Question

Q10. Solve for x, y, z and t , if

$$2 \begin{bmatrix} x & y \\ z & t \end{bmatrix} + \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 7 & 10 \\ 11 & 14 \end{bmatrix}$$

Solution

Equating corresponding elements:

$$x = 3, \quad y = 4, \quad z = 4, \quad t = 5$$

Question

Q11. If

$$x \begin{bmatrix} 3 \\ 2 \end{bmatrix} + y \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 11 \\ 1 \end{bmatrix},$$

find x and y .

Solution

$$3x + y = 11, \quad 2x - y = 1$$

Solving,

$$x = \frac{12}{5}, \quad y = \frac{19}{5}$$

Question

Q12. Find x, y, z and w , if

$$2 \begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} 4 & 6 \\ 2 & 8 \end{bmatrix} + \begin{bmatrix} 2 & y \\ z & 0 \end{bmatrix}$$

Solution

Comparing corresponding elements:

$$x = 3, \quad y = 6, \quad z = 2, \quad w = 4$$

Transpose of a Matrix

Definition

Let $A = [a_{ij}]$ be an $m \times n$ matrix. The matrix obtained by interchanging the rows and columns of A is called the **transpose of A** and is denoted by A' or A^T .

If $A = [a_{ij}]_{m \times n}$, then

$$A' = [a_{ji}]_{n \times m}.$$

Example

If

$$A = \begin{bmatrix} 3 & 5 \\ \sqrt{3} & 1 \\ 0 & -1 \end{bmatrix},$$

then

$$A' = \begin{bmatrix} 3 & \sqrt{3} & 0 \\ 5 & 1 & -1 \end{bmatrix}.$$

1.1 Properties of Transpose of Matrices

Definition

For any matrices A and B of suitable orders and scalar k :

- (i) $(A')' = A$
- (ii) $(kA)' = kA'$
- (iii) $(A + B)' = A' + B'$
- (iv) $(AB)' = B'A'$

Example

Let

$$A = \begin{bmatrix} 3 & \sqrt{3} & 2 \\ 4 & 2 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & -1 & 2 \\ 1 & 2 & 4 \end{bmatrix}.$$

Verify:

- (i) $(A')' = A$
- (ii) $(A + B)' = A' + B'$
- (iii) $(kB)' = kB'$

Solution

(i)

$$A' = \begin{bmatrix} 3 & 4 \\ \sqrt{3} & 2 \\ 2 & 0 \end{bmatrix} \Rightarrow (A')' = A$$

(ii)

$$A + B = \begin{bmatrix} 5 & \sqrt{3} - 1 & 4 \\ 5 & 4 & 4 \end{bmatrix} \Rightarrow (A + B)' = \begin{bmatrix} 5 & 5 \\ \sqrt{3} - 1 & 4 \\ 4 & 4 \end{bmatrix}$$

$$A' + B' = \begin{bmatrix} 5 & 5 \\ \sqrt{3} - 1 & 4 \\ 4 & 4 \end{bmatrix}$$

Hence, $(A + B)' = A' + B'$.

(iii)

$$(kB)' = kB'$$

Symmetric and Skew-Symmetric Matrices

Definition

A square matrix $A = [a_{ij}]$ is said to be **symmetric** if

$$A' = A \quad \text{or} \quad a_{ij} = a_{ji}.$$

Example

$$A = \begin{bmatrix} \sqrt{3} & 2 & 3 \\ 2 & -1.5 & -1 \\ 3 & -1 & 1 \end{bmatrix}$$

is a symmetric matrix since $A' = A$.

Definition

A square matrix $A = [a_{ij}]$ is said to be **skew-symmetric** if

$$A' = -A \quad \text{or} \quad a_{ij} = -a_{ji}.$$

All diagonal elements of a skew-symmetric matrix are zero.

Example

$$B = \begin{bmatrix} 0 & e & f \\ -e & 0 & g \\ -f & -g & 0 \end{bmatrix}$$

is a skew-symmetric matrix since $B' = -B$.

Theorems

Definition

[Theorem 1] For any square matrix A with real entries:

$A + A'$ is symmetric and $A - A'$ is skew-symmetric.

Solution

Let $B = A + A'$.

$$B' = (A + A')' = A' + (A')' = A' + A = B$$

Hence, B is symmetric.

Similarly, let $C = A - A'$.

$$C' = (A - A')' = A' - A = -C$$

Hence, C is skew-symmetric.

Definition

[Theorem 2] Any square matrix A can be expressed as the sum of a symmetric and a skew-symmetric matrix.

Solution

$$A = \frac{1}{2}(A + A') + \frac{1}{2}(A - A')$$

Here,

$\frac{1}{2}(A + A')$ is symmetric, and $\frac{1}{2}(A - A')$ is skew-symmetric.

Example

Express the matrix

$$B = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$$

as the sum of a symmetric and a skew-symmetric matrix.

Solution

First, find the transpose of B .

$$B' = \begin{bmatrix} 2 & -1 & 1 \\ -2 & 3 & -2 \\ -4 & 4 & -3 \end{bmatrix}$$

Step 1: Symmetric part

$$P = \frac{1}{2}(B + B')$$

$$P = \frac{1}{2} \begin{bmatrix} 4 & -3 & -3 \\ -3 & 6 & 2 \\ -3 & 2 & -6 \end{bmatrix} = \begin{bmatrix} 2 & -\frac{3}{2} & -\frac{3}{2} \\ -\frac{3}{2} & 3 & 1 \\ -\frac{3}{2} & 1 & -3 \end{bmatrix}$$

Since $P' = P$, P is a symmetric matrix.

Step 2: Skew-symmetric part

$$Q = \frac{1}{2}(B - B')$$

$$Q = \frac{1}{2} \begin{bmatrix} 0 & -1 & -5 \\ 1 & 0 & 6 \\ 5 & -6 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -\frac{1}{2} & -\frac{5}{2} \\ \frac{1}{2} & 0 & 3 \\ \frac{5}{2} & -3 & 0 \end{bmatrix}$$

Since $Q' = -Q$, Q is a skew-symmetric matrix.

Final Result:

$$B = P + Q$$

Hence, the matrix B is expressed as the sum of a symmetric and a skew-symmetric matrix.

Invertible Matrices

Definition

Let A be a square matrix of order m . If there exists another square matrix B of the same order such that

$$AB = BA = I,$$

then B is called the **inverse** of A and is denoted by A^{-1} . In this case, the matrix A is said to be **invertible**.

Example

Let

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix}.$$

Show that B is the inverse of A .

Solution

We compute

$$AB = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 4-3 & -6+6 \\ 2-2 & -3+4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I.$$

Similarly,

$$BA = I.$$

Hence, $B = A^{-1}$ and A is invertible.

Note

1. A rectangular matrix does **not** possess an inverse.
2. If B is the inverse of A , then A is also the inverse of B .

Theorem 3 (Uniqueness of Inverse)

Theorem

If the inverse of a square matrix exists, then it is unique.

Solution

Let A be a square matrix of order m . Suppose B and C are two inverses of A . Then,

$$AB = BA = I \quad \text{and} \quad AC = CA = I.$$

Now,

$$B = BI = B(AC) = (BA)C = IC = C.$$

Hence, $B = C$. Therefore, the inverse of a matrix, if it exists, is unique.

Theorem 4

Theorem

If A and B are invertible matrices of the same order, then

$$(AB)^{-1} = B^{-1}A^{-1}.$$

Solution

We know that

$$AB(AB)^{-1} = I.$$

Pre-multiplying both sides by A^{-1} , we get

$$B(AB)^{-1} = A^{-1}.$$

Pre-multiplying both sides by B^{-1} ,

$$(AB)^{-1} = B^{-1}A^{-1}.$$

Hence proved.

Solved Questions on Matrices

Question

Q1. Find the order of the matrix

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 4 & 0 & -1 \end{bmatrix}.$$

Solution

The given matrix has 2 rows and 3 columns.

$$\text{Order of } A = 2 \times 3.$$

Question

Q2. If

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix},$$

find $A + B$.

Solution

$$A + B = \begin{bmatrix} 1+5 & 2+6 \\ 3+7 & 4+8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}.$$

Question

Q3. Find $2A$, where

$$A = \begin{bmatrix} 3 & -1 \\ 0 & 4 \end{bmatrix}.$$

Solution

$$2A = \begin{bmatrix} 6 & -2 \\ 0 & 8 \end{bmatrix}.$$

Question

Q4. If

$$A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix},$$

find $A - B$.

Solution

$$A - B = \begin{bmatrix} 2-1 & 1-0 \\ 4-(-1) & 3-2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 5 & 1 \end{bmatrix}.$$

Question

Q5. Find AB , where

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}.$$

Solution

$$AB = \begin{bmatrix} 1(2) + 2(1) & 1(0) + 2(1) \\ 3(2) + 4(1) & 3(0) + 4(1) \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ 10 & 4 \end{bmatrix}.$$

Question

Q6. Find the transpose of

$$A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}.$$

Solution

$$A^T = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}.$$

Question

Q7. Verify that $(A^T)^T = A$ for

$$A = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}.$$

Solution

$$A^T = \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix} \Rightarrow (A^T)^T = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix} = A.$$

Hence verified.

Question

Q8. Show that $(A + B)^T = A^T + B^T$ for

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}.$$

Solution

$$(A + B)^T = \begin{bmatrix} 6 & 10 \\ 8 & 12 \end{bmatrix} = A^T + B^T.$$

Question

Q9. Determine whether the matrix

$$A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

is symmetric.

Solution

$$A^T = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = A.$$

Hence, A is symmetric.

Question

Q10. Find the inverse of

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

Solution

$$|A| = (1)(4) - (2)(3) = -2 \neq 0.$$

$$A^{-1} = \frac{1}{-2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}.$$

Question

Q11. Verify that $AB \neq BA$ for

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}.$$

Solution

$$AB = \begin{bmatrix} 4 & 6 \\ 1 & 3 \end{bmatrix}, \quad BA = \begin{bmatrix} 2 & 4 \\ 1 & 5 \end{bmatrix}.$$

Thus, $AB \neq BA$.

Question

Q12. Express the matrix

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

as the sum of a symmetric and a skew-symmetric matrix.

Solution

$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T).$$

$$\frac{1}{2}(A + A^T) = \begin{bmatrix} 1 & \frac{5}{2} \\ \frac{5}{2} & 4 \end{bmatrix}, \quad \frac{1}{2}(A - A^T) = \begin{bmatrix} 0 & -\frac{1}{2} \\ \frac{1}{2} & 0 \end{bmatrix}.$$

Exercise : Matrices

Question

Q1. Find the order of the matrix

$$A = \begin{bmatrix} 3 & 0 & -2 \\ 1 & 4 & 5 \end{bmatrix}.$$

Question

Q2. If

$$A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix},$$

find $A + B$.

Question

Q3. Find $3A$, where

$$A = \begin{bmatrix} -1 & 2 \\ 4 & 0 \end{bmatrix}.$$

Question

Q4. Find $A - B$, where

$$A = \begin{bmatrix} 5 & 2 \\ 1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}.$$

Question

Q5. If

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 1 & 4 \end{bmatrix},$$

find AB .

Question

Q6. Find the transpose of the matrix

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 0 & 5 \end{bmatrix}.$$

Question

Q7. Verify whether the matrix

$$A = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$$

is symmetric.

Question

Q8. Determine whether the matrix

$$A = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$$

is skew-symmetric.

Question

Q9. Find the inverse of the matrix

$$A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix},$$

if it exists.

Question

Q10. Solve for x and y :

$$\begin{bmatrix} x & 2 \\ 3 & y \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix}.$$

Question

Q11. Express the matrix

$$A = \begin{bmatrix} 3 & 1 \\ 5 & 7 \end{bmatrix}$$

as the sum of a symmetric and a skew-symmetric matrix.

Question

Q12. If

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix},$$

verify whether $AB = BA$.

Answer List

Result

(1) 2×3

(2) $\begin{bmatrix} 3 & 0 \\ 3 & 6 \end{bmatrix}$

(3) $\begin{bmatrix} -3 & 6 \\ 12 & 0 \end{bmatrix}$

(4) $\begin{bmatrix} 3 & 1 \\ 2 & 3 \end{bmatrix}$

(5) $\begin{bmatrix} 4 & 8 \\ 7 & 4 \end{bmatrix}$

(6) $\begin{bmatrix} 2 & 4 \\ -1 & 0 \\ 3 & 5 \end{bmatrix}$

(7) Symmetric matrix

(8) Skew-symmetric matrix

(9) $A^{-1} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$

(10) $x = 4, y = 1$

(11) $\frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$

(12) $AB \neq BA$